

SOFTWARE AND HARDWARE TOOLS

1. Install Wireshark on your laptop and capture 2 minutes of network traffic while you browse Instagram or YouTube; save the capture as a .pcap file and note the top 3 protocols you see.

Steps

- 1. Install **Wireshark**.
- 2. Start capturing on your **Wi-Fi** or **Ethernet** interface.
- 3. Browse **Instagram** or **YouTube** for **2 minutes**.
- 4. Stop the capture.
- 5. Save it as **network_capture.pcap**.

Top 3 Protocols (Example)

Protocol	Purpose
TLS	Secure HTTPS communication
TCP	Reliable data transfer
DNS	Converts website names to IP addresses

2. Use the 'ping' command in Command Prompt or Terminal to test connectivity to www.spotify.com and www.zomato.com; record the response times and identify which site responds faster.

```
C:\Windows\System32>ping www.spotify.com

Pinging atc.spotify.map.fastly.net [2a04:4e42:31::810] with 32 bytes of data:
Reply from 2a04:4e42:31::810: time=44ms
Reply from 2a04:4e42:31::810: time=30ms
Reply from 2a04:4e42:31::810: time=32ms
Reply from 2a04:4e42:31::810: time=25ms

Ping statistics for 2a04:4e42:31::810:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 25ms, Maximum = 44ms, Average = 32ms

C:\Windows\System32>
C:\Windows\System32>ping www.zomato.com

Pinging e70929.dsca.akamaiedge.net [2600:140f:7200:49::17d4:73] with 32 bytes of data:
Reply from 2600:140f:7200:49::17d4:73: time=44ms
Reply from 2600:140f:7200:49::17d4:73: time=46ms
Reply from 2600:140f:7200:49::17d4:73: time=46ms
Reply from 2600:140f:7200:49::17d4:73: time=38ms

Ping statistics for 2600:140f:7200:49::17d4:73:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 38ms, Maximum = 46ms, Average = 43ms

C:\Windows\System32>
```

Faster Website: Spotify (example only—your results may differ).

3. Open Task Manager (Windows) or Activity Monitor (Mac) and list all running processes related to Chrome, WhatsApp Desktop, or any browser-based app you use; note the process name and memory usage for each.

Windows

1. Press Ctrl + Shift + Esc
2. Open Task Manager
3. Go to Processes

Example

Process Name Memory Usage

Chrome.exe 520 MB

Chrome.exe 120 MB

Process Name Memory Usage

WhatsApp.exe 180 MB

**4. Download and run the free version of 'Angry IP Scanner' or 'Advanced IP Scanner' to scan your local WiFi network; list the IP and device name of at least 2 devices connected to your network.

Hint: Make sure your firewall allows scanning, and only scan your own home/college network.**

Tool

- Advanced IP Scanner or Angry IP Scanner

Example Output

IP Address	Device Name
192.168.1.1	TP-Link Router
192.168.1.100	Hensie-Laptop

5. Use ChatGPT or Bing AI to suggest 3 hardware tools (like cable tester, crimping tool, or multimeter) used in network troubleshooting; for each, write one line describing when you would use it.

Tool	When It Is Used
Cable Tester	Checks whether an Ethernet cable is wired correctly and detects faults.
RJ-45 Crimping Tool	Used to attach RJ-45 connectors to Ethernet cables.
Digital Multimeter	Measures voltage and checks power supply or cable continuity during troubleshooting.